

Modular Long-Context Sequence Modeling on WikiText-2 Attention Mechanisms, Positional Encodings, Hybrid Architectures, and AFT Evaluation

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1 Introduction

This project investigates efficient long-context sequence modeling under strict hardware constraints using the WikiText-2 dataset. Starting from a baseline decoder-only Transformer, the study systematically evaluates alternative attention mechanisms, positional encoding strategies, convolution-attention hybrid architectures, and Attention-Free Transformer (AFT) variants. The primary objective is to understand the trade-offs between language modeling performance and computational efficiency. All experiments are conducted within an 8GB VRAM budget while tracking validation perplexity, memory consumption, throughput, and inference latency. The implementation follows a modular design in which attention layers, positional encodings, and architectural blocks can be independently exchanged, enabling controlled comparison between different long-context modeling approaches.

2 Experimental Setup

2.1 Hardware Profile and Resource Boundaries

All experiments were executed locally on Ubuntu Linux through terminal-based PyTorch training workflows utilizing a dedicated **NVIDIA RTX 4060 GPU (8GB VRAM)**. The available VRAM budget served as the primary design constraint governing model width, depth, batch sizing, context length selection, and optimization strategy.

2.2 Memory Sourcing and Telemetry Disclaimer

Peak GPU memory usage values throughout this report were obtained directly from the automatically logged Weights & Biases system monitoring graph (**GPU Memory Allocated**), which records the actual physical VRAM allocation during execution. Although a custom code metric (`perf/peak_gpu_mem_MB`) was originally set up in the script, it incorrectly remained constant across different context lengths due to a logging bug. Therefore, the built-in Weights & Biases system panel was used exclusively for accurate memory analysis across all tasks.

2.3 Dataset and Tokenization Pipeline

The WikiText-2 corpus was loaded directly from the HuggingFace Datasets library and converted into continuous token streams. Tokenization was performed using the GPT-2 Byte Pair Encoding (BPE) tokenizer provided by `tiktoken`.

Table 1: Dataset and Tokenization Specifications

Parameter	Value
Vocabulary Size	50,257 tokens
Tokenizer	GPT-2 BPE (<code>tiktoken</code>)
Training Objective	Next-token prediction
Context Length	Dynamically varied per experiment

2.4 Optimization and Training Configuration

To ensure fair comparison between architectural variants, all experiments used a standardized optimization pipeline.

Table 2: Optimization and Training Hyperparameters

Hyperparameter	Value
Optimizer	AdamW
Peak Learning Rate	5×10^{-4}
Minimum Learning Rate	1×10^{-6}
Weight Decay	0.01
Epoch Budget	5 epochs
Gradient Clipping	1.0
Mixed Precision Training	FP16 Automatic Mixed Precision (AMP)
Batch Size	2 sequences/GPU
Gradient Accumulation	4 steps
Effective Batch Size	8 sequences

Learning rate scheduling followed a two-stage policy:

1. Linear warmup during the first 10% of optimization steps.
2. Cosine decay from peak learning rate to the minimum learning rate for the remainder of training.

To remain within the 8GB memory budget at long context lengths, activation checkpointing was enabled across Transformer blocks, significantly reducing activation storage during backpropagation.

3 Task 1: Baseline Transformer

The baseline model implements a standard decoder-only GPT-2 style Transformer using full causal Multi-Head Self-Attention (MHA) and fixed sinusoidal positional encodings. This con-

figuration serves as the reference architecture against which all attention, positional encoding, hybrid convolution, and AFT variants are compared.

3.1 Architecture Specification

Table 3: Baseline Transformer Configuration

Parameter	Value
Vocabulary Size	50,257
Embedding Dimension (d_{model})	768
Transformer Layers	6
Attention Heads	12
Head Dimension	64
Feedforward Expansion Ratio	$4\times$
FFN Hidden Dimension	3072
Dropout Rate	0.1
QKV Bias	Disabled
Positional Encoding	Sinusoidal
Attention Type	Multi-Head Attention (MHA)

3.2 Memory-Constrained Design Choices

To maximize representational capacity while remaining strictly executable on an 8GB consumer GPU resource boundary, several critical architecture scaling choices were implemented. The overall Transformer depth was scaled down from 12 to 6 layers, and training was executed using FP16 Automatic Mixed Precision (AMP) to preserve memory bandwidth. Furthermore, activation checkpointing was systematically applied to all Transformer blocks to reduce cached storage during the backward pass, while gradient accumulation was utilized across 4 steps to scale up the effective batch size. Finally, a dynamic context length configuration was leveraged across experimental runs to isolate sequence boundaries safely within hardware limits.

3.3 Baseline Performance

The baseline model configuration was fully trained and verified to establish foundational downstream cross-entropy and tracking reference boundaries. Table 4 details the empirical baseline results captured at a standard context window of 1024 tokens.

Table 4: Baseline Transformer Performance Metrics

Metric / Parameter	Value
Attention Variant	Causal MHA
Positional Variant	Sinusoidal
Training Context Length (L_{train})	1024 tokens
Training Loss	3.808
Validation Loss	4.305
Validation Perplexity (PPL)	74.129
Peak GPU Memory Usage	4048 MB
Training Time per Epoch	310 sec
Training Throughput	30524 tokens/sec

Baseline Weights & Biases Run Workspace Link:

Baseline W&B run

4 Task 2: Attention Variants

4.1 Architectural Mechanics

To solve the speed and memory issues of standard attention, we tested three efficient attention variants:

- **Grouped Query Attention (GQA):** Groups query heads together so they can share the same key and value tracks. With H query heads and G groups, the group multiplier is $S = H/G$. This slashes the size of the KV cache stored in memory:

$$\mathbf{K}_{\text{expanded}} = \text{repeat_interleave}(\mathbf{K}_{\text{group}}, S, \text{dim}=1) \quad (1)$$

$$\mathbf{V}_{\text{expanded}} = \text{repeat_interleave}(\mathbf{V}_{\text{group}}, S, \text{dim}=1) \quad (2)$$

- **Sliding Window Attention (SWA):** Limits how far back the model can look by setting a strict window size (W). This drops the compute required for long inputs down to a simple linear cost:

$$\mathbf{D}_{t,t'} = \text{pos}_{\text{query},t} - \text{pos}_{\text{key},t'} \quad (3)$$

$$\mathbf{M}_{t,t'} = (\mathbf{D}_{t,t'} < 0) \vee (\mathbf{D}_{t,t'} \geq W) \quad (4)$$

- **Linear Attention (Kernelized):** Inverts the matrix multiplication order using a feature map $\phi(\cdot)$ to make the model run much faster on long sequences:

$$\phi(x) = \text{clamp_min}(\text{ELU}(x) + 1.0, 10^{-4}) \quad (5)$$

$$\mathbf{Y}_t = \frac{\phi(\mathbf{Q}_t) \sum_{t'=1}^t \phi(\mathbf{K}_{t'})^\top \mathbf{V}_{t'}}{\phi(\mathbf{Q}_t) \sum_{t'=1}^t \phi(\mathbf{K}_{t'})^\top} \quad (6)$$

This version does not work with relative position biases and will crash with a `ValueError` if combined.

4.2 Comparative Analysis

The results across different context lengths show clear trade-offs between the variants. At the max context length of 2048 tokens, Sliding Window Attention (SWA) gets the lowest validation perplexity (88.65). This shows it is highly effective at keeping local context while safely throwing away old, unneeded tokens. Grouped Query Attention (GQA) is the most balanced variant for hardware resources, using the least peak memory (5728 MB) and running the fastest (24,592 tokens/sec) due to its shared key-value head layout.

On the other hand, Linear Attention performs poorly in vanilla PyTorch. Its perplexity drops significantly (107.46) and it uses a lot of memory (6516 MB) at the 2048 limit. This happens because calculating cumulative sums (`torch.cumsum`) creates a massive memory bottleneck without custom CUDA or Triton kernels. Therefore, SWA was selected as the baseline for the next tasks.

Table 5: Attention Variant Multi-Context Performance Matrix (Sinusoidal Encodings)

Context Len (L)	Attention Model	Train Loss	Val Loss	Val PPL $\downarrow$	Peak VRAM (MB/GB) $\downarrow$	Epoch Time (s)	Throughput (tok/sec) $\uparrow$
512	Sliding Window (SWA)	3.04	4.18	65.43	3368 MB	240	30604
	Grouped Query (GQA)	3.16	4.18	65.42	3186 MB	233	31662
	Linear (Kernelized)	3.41	4.35	77.45	3720 MB	555	15691
1024	Sliding Window (SWA)	3.78	4.28	72.52	4069 MB	300	31081
	Grouped Query (GQA)	3.69	4.32	74.92	3673 MB	295	31709
	Linear (Kernelized)	4.10	4.49	89.35	4050 MB	569	17299
2048	Sliding Window (SWA)	4.35	4.48	88.65	5755 MB	456	24200
	Grouped Query (GQA)	4.41	4.55	95.04	5728 MB	456	24592
	Linear (Kernelized)	4.55	4.68	107.46	6516 MB	614	18017

Weights & Biases Attention Variant Workspace Run Logs:

- **Context Length 512:** SWA (L=512) Run · GQA (L=512) Run · Linear Attention (L=512) Run
- **Context Length 1024:** SWA (L=1024) Run · GQA (L=1024) Run · Linear Attention (L=1024) Run
- **Context Length 2048:** SWA (L=2048) Run · GQA (L=2048) Run · Linear Attention (L=2048) Run

5 Task 3: Positional Encodings & Length Extrapolation Test

This section tests how well the model can handle longer text sequences at test time than it saw during training. Models were trained on short sequences ($L_{\text{train}} = 512$) and evaluated on longer horizons ($L_{\text{test}} \in \{512, 1024, 2048\}$) without any extra tuning.

5.1 Positional Modeling Frameworks

We integrated three different methods to handle token positions:

- **Rotary Positional Embedding (RoPE):** Adds position info by rotating the Query and Key vectors, which helps capture distances between tokens relatively:

$$\mathbf{R}_{\theta_i, m}^2 \mathbf{x} = \begin{pmatrix} \cos m\theta_i & -\sin m\theta_i \\ \sin m\theta_i & \cos m\theta_i \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \quad (7)$$

$$\mathbf{q}_{\text{rope}} = (\mathbf{q} \odot \mathbf{cos}) + (\text{neg_half}(\mathbf{q}) \odot \mathbf{sin}) \quad (8)$$

- **Attention with Linear Biases (ALiBi):** Instead of using position vectors at the start, it injects a fixed negative penalty to the attention scores based on how far apart the tokens are:

$$\text{Attention}_b(Q, K) = \text{Softmax} \left(\frac{\mathbf{QK}^\top}{\sqrt{d_{\text{head}}}} + m \cdot \mathbf{A} \right) \quad (9)$$

Where $\mathbf{A}_{t, t'} = t' - t$ for $t' \leq t$, and the slopes m are mathematically fixed per head.

- **Relative Positional Encoding:** Learns specific embeddings for the exact distance between two tokens, rather than focusing on their absolute positions in the text:

$$\text{Attention}_{t, t'} = \frac{\mathbf{Q}_t \mathbf{K}_{t'}^\top + \mathbf{Q}_t \mathbf{R}_{\psi(t-t')}^\top}{\sqrt{d_{\text{head}}}} \quad (10)$$

Where $\mathbf{R}$ is a learned dictionary of relative position gaps clipped to a fixed window.

5.2 Comparative Analysis

The length extrapolation experiment shows a clear split between RoPE and the distance-biased frameworks when text sequences grow longer. When trained at $L_{\text{train}} = 512$, the Rotary Position Embedding (RoPE) model fails completely on longer text, with its validation perplexity jumping from 50.96 up to 63.19 at $L_{\text{test}} = 2048$. This happens because the model struggles with new rotation angles it did not see during training.

In contrast, both ALiBi and Relative Positional Encoding handle context expansion easily without any fine-tuning. Their perplexity actually improves as the context window widens (ALiBi drops down to 50.20, and Relative Position drops down to 48.99 at $L_{\text{test}} = 2048$). This proves that fixed distance penalties help the model use longer context naturally. Relative Positional Encoding was chosen as our best variant since it gives the lowest training loss (2.69) and the best validation perplexity.

Table 6: Positional Encoding Training Performance Profiles at $L_{\text{train}} = 512$

Positional Variant	Train Loss	Val Loss	Val PPL ↓	Peak VRAM (MB/GB) ↓	Epoch Time (s)	Throughput (tok/sec) ↑
RoPE	2.83	3.93	51.09	3197 MB	249	29679
ALiBi	2.86	4.00	55.12	3210 MB	262	28805
Relative Position Encoding	2.69	3.95	52.02	3220 MB	304	25550

Weights & Biases Positional Variant Training Run Logs ($L_{\text{train}} = 512$):

RoPE Training Run · ALiBi Training Run · Relative Position Encoding Training Run

Table 7: Length Extrapolation Evaluation Matrix (Validation Perplexity)

Positional Variant	PPL @ $L_{\text{test}} = 512$	PPL @ $L_{\text{test}} = 1024$	PPL @ $L_{\text{test}} = 2048$
RoPE	50.96	54.65	63.19
ALiBi	54.89	52.25	50.20
Relative Position Encoding	51.52	49.44	48.99

6 Task 4: Convolution + Attention Hybrids (Conformer Architecture)

To help the model look at local neighbor words better, we combined 1D convolutions with the global attention layer, inspired by the Conformer architecture. This allows the model to catch tight local patterns alongside long-range relationships.

6.1 Causal Hybrid Formulations and Topologies

To capture local features safely without leaking future words, the model uses **Causal 1D Depthwise Separable Convolutions** with left-padding (kernel_size=1). The **CausalConvModule** runs inputs through layer normalization, channel expansion ($d_{\text{model}} \rightarrow 2d_{\text{model}}$), a Gated Linear Unit (GLU) activation, causal 1D convolution, and a residual add:

$$\mathbf{X}_{\text{norm}} = \text{LayerNorm}(\mathbf{X}) \quad (11)$$

$$\mathbf{X}_{\text{expand}} = \mathbf{X}_{\text{norm}} \mathbf{W}_{\text{expand}} \quad (12)$$

$$\mathbf{X}_{\text{glu}} = \mathbf{X}_1 \odot \sigma(\mathbf{X}_2) \quad (13)$$

$$\mathbf{X}_{\text{conv}} = \text{CausalDepthwiseConv1d}(\mathbf{X}_{\text{glu}}) \quad (14)$$

$$\mathbf{X}_{\text{act}} = \text{SiLU}(\text{LayerNorm}(\mathbf{X}_{\text{conv}})) \quad (15)$$

$$\mathbf{X}_{\text{proj}} = \text{Dropout}(\mathbf{X}_{\text{act}} \mathbf{W}_{\text{project}}) \quad (16)$$

$$\mathbf{Y} = \mathbf{X} + \mathbf{X}_{\text{proj}} \quad (17)$$

We designed and tested two distinct layout topologies at a context limit of 2048 tokens:

- **Design A: Interleaved Convolution + Attention** places the convolution block right after global attention and before the standard feedforward network (FFN):

$$\mathbf{X}^{(1)} = \mathbf{X} + \text{Dropout}(\text{Attn}(\text{LayerNorm}(\mathbf{X}))) \quad (18)$$

$$\mathbf{X}^{(2)} = \text{CausalConvModule}(\mathbf{X}^{(1)}) \quad (19)$$

$$\mathbf{Y} = \mathbf{X}^{(2)} + \text{Dropout}(\text{FFN}(\text{LayerNorm}(\mathbf{X}^{(2)}))) \quad (20)$$

- **Design B: Gated Convolutional FFN** completely replaces the standard FFN block with our **CausalConvModule**, using the internal GLU layer instead of regular MLP layers:

$$\mathbf{X}^{(1)} = \mathbf{X} + \text{Dropout}(\text{Attn}(\text{LayerNorm}(\mathbf{X}))) \quad (21)$$

$$\mathbf{Y} = \text{CausalConvModule}(\mathbf{X}^{(1)}) \quad (22)$$

Critical Architectural Note (Caching Constraint):

Because 1D convolution layers need to see the entire text window to calculate features correctly, standard token-by-token KV caching does not work out-of-the-box. We set `use_cache=False` to force full context processing during generation steps.

6.2 Comparative Analysis

We combined our best attention variant (SWA) and best positional encoding (Relative Position) to test the hybrid designs at a context length of 2048. The Interleaved Convolution + Attention design performs the best, reaching the lowest validation perplexity (**60.74 PPL**). This proves that placing a local convolution layer right after global attention helps the model capture text structure much better than standard attention alone.

Looking at the compute metrics, the Gated Convolutional FFN has a slightly higher perplexity (72.56) but saves about 37% of the parameters compared to a standard dense FFN. This cuts down epoch training times (706 s) and makes it run much faster (15,928 tokens/sec), making it an excellent lightweight option if you are running low on VRAM.

Table 8: Hybrid Architectural Profiles at $L_{\text{train}} = 2048$

Hybrid Design Layout	Train Loss	Val Loss	Val PPL ↓	Peak VRAM (MB) ↓	Epoch Time (s)	Throughput (tok/sec) ↑
Gated Convolutional FFN	4.10	4.28	72.56	5791	706	15928
Interleaved Conv + Attention	3.78	4.10	60.74	5926	808	13880
Best Attention + Positional Variant (SWA + Rel)	4.35	4.48	88.65	5755	456	24200

Weights & Biases Hybrid Variant Workspace Run Logs ($L_{\text{train}} = 2048$):

Gated Convolutional FFN Run · Interleaved Conv + Attention Run · SWA + Relative Position Baseline Run

7 Task 5: BONUS — The Attention-Free Transformer (AFT)

7.1 AFT Variant Formulations and Stability Mechanics

The Attention-Free Transformer (AFT) gets rid of the heavy $N \times N$ attention matrix entirely. It replaces it with a query-gated pooling setup that uses way less memory on long sequences. We implemented four versions:

1. **AFT-Simple (AFTSimple)**: Uses no position biases and enforces causality using simple cumulative sums (`torch.cumsum`) along the sequence dimension:

$$\mathbf{Y}_t = \sigma(\mathbf{Q}_t) \odot \frac{\sum_{t'=1}^t \exp(\mathbf{K}_{t'}) \odot \mathbf{V}_{t'}}{\sum_{t'=1}^t \exp(\mathbf{K}_{t'})} \quad (23)$$

2. **AFT-Full (AFTFull)**: Introduces learned absolute position biases to create a full time-bias matrix:

$$\mathbf{Y}_t = \sigma(\mathbf{Q}_t) \odot \frac{\sum_{t'=1}^t \exp(\mathbf{W}_{t,t'} + \mathbf{K}_{t'}) \odot \mathbf{V}_{t'}}{\sum_{t'=1}^t \exp(\mathbf{W}_{t,t'} + \mathbf{K}_{t'})} \quad (24)$$

3. **AFT-Local (AFTLocal)**: Limits the learned position biases to a fixed local window size (W_{local}):

$$\mathbf{W}_{t,t'} = \begin{cases} [\mathbf{U}\mathbf{V}^\top]_{t,t'} & \text{if } |t - t'| < W_{\text{local}} \\ 0.0 & \text{if } |t - t'| \geq W_{\text{local}} \end{cases} \quad (25)$$

4. **AFT-Conv (AFTConv)**: Mixes global context with local 1D convolutions grouped by attention heads to balance local and global features:

$$\mathbf{w}_{\text{norm}} = \frac{\mathbf{w} - \mu_{\mathbf{w}}}{\sqrt{\sigma_{\mathbf{w}}^2 + 10^{-6}}} \quad (26)$$

$$\text{Filter} = \exp(\gamma \cdot \mathbf{w}_{\text{norm}} + \beta) - 1.0 \quad (27)$$

To stop the model from crashing due to huge exponential numbers, all versions run in stable full `float32` precision (turning off mixed precision) and subtract the maximum key value before calculating exponents:

$$\mathbf{K}_{\text{stable}} = \mathbf{K} - \max_{1 \leq t \leq T}(\mathbf{K}_t) \quad (28)$$

7.2 Comparative Analysis

Comparing the Attention-Free Transformer (AFT) variants against our best hybrid model (Interleaved Conv + Attention) shows a major trade-off between speed and accuracy. AFT-Simple is incredibly fast, reaching a throughput of 45,709 tokens/sec and using the least memory (5397 MB), but it suffers from being position-blind, which leads to a high perplexity of 98.42.

Adding local convolutions in AFT-Conv fixes this tracking problem, keeping throughput very high (46,467 tokens/sec) while managing text patterns better. However, none of the attention-free models can match the accuracy of the Interleaved Hybrid model (60.74 PPL). This shows that standard dot-product attention is still much better at capturing complex, long-range language relationships than attention-free alternatives.

Table 9: AFT Model Variants vs. Interleaved Hybrid Baseline at Context Length 2048

Model Framework Variant	Train Loss	Val Loss	Val PPL ↓	Peak VRAM (MB) ↓	Epoch Time (s)	Throughput (tok/sec) ↑
Task 4: Interleaved Hybrid	3.78	4.10	60.74	5926	808	13880
Bonus: AFT-Simple	4.42	4.59	98.42	5397	236	45709
Bonus: AFT-Full	4.49	4.58	97.66	5440	305	35736
Bonus: AFT-Local	4.48	4.58	97.54	5411	308	35481
Bonus: AFT-Conv	4.47	4.61	100.54	5285	232	46467

Weights & Biases AFT Evaluation Workspace Run Logs ($L_{\text{train}} = 2048$):

Task 4: Interleaved Hybrid Baseline Run · Bonus: AFT-Simple Run · Bonus: AFT-Full Run · Bonus: AFT-Local Run · Bonus: AFT-Conv Run

8 Limitations

Due to strict time and compute constraints, all experimental model configurations were trained for a limited budget of only 5 epochs. This truncated training window implies that complete convergence may not have been fully attained across all runs, as optimization concluded before the validation loss could reach its absolute minimum. Consequently, the empirical metrics and findings presented throughout this report should be interpreted strictly as a comparative scaling analysis under identical hardware constraints rather than indicators of absolute optimal performance. Additionally, the local 8GB VRAM boundary heavily dictated the model’s design choices, acting as a strict bottleneck that capped overall layer depth and prevented the exploration of larger hyperparameter spaces or wider architectural frameworks.

9 Conclusion

This project evaluated how different attention types, positional encodings, and hybrid blocks perform under tight memory limits. Our main findings are:

1. **Best Long-Context Setup:** Tests show that combining **Sliding Window Attention (SWA)** with **Relative Positional Encoding** works best for long sequences. SWA keeps things fast by focusing only on nearby words, while Relative Positional Encoding lets the model smoothly extrapolate to longer text sizes up to 2048 tokens without losing accuracy.
2. **Power of Hybrid Layers:** Adding a causal 1D convolution layer right after global attention gives the highest accuracy boost. The **Interleaved Convolution + Attention** model achieved the lowest overall validation perplexity (**60.74 PPL**), proving that mixing local n-gram filters with windowed global attention speeds up language learning.
3. **Speed vs. Accuracy Trade-offs:** Attention-free methods like **Gated Convolutional FFN** and **AFT-Conv** offer massive speedups. AFT-Conv hits an outstanding throughput of **46,467 tokens/sec**, and the Gated FFN layout cuts active block parameter counts by $\sim 37\%$. However, they still trail behind standard multi-head attention in terms of perplexity. For resource-constrained setups, combining local convolutions with windowed attention and relative encodings provides the best balance of speed and modeling accuracy.

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